LAB 9: Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

Objectives

1. To explore the working principles of a Wireless Local Area Network (WLAN) and gain practical experience in configuring a Wireless Router.

2. To configure the Service Set Identifier (SSID) and apply WPA2 security protocols in order to protect the wireless network from unauthorised access.

3. To establish wireless connectivity between client Laptops and the Wireless Router, and validate the connection using packet analysis through the Ping utility.

Theory

1. WLAN (Wireless Local Area Network)

A WLAN is a type of computer network that interconnects two or more devices through wireless communication, forming a local area network (LAN) within a restricted area such as a home, school, or office building.

Operation: It transmits data between devices and a wireless router or access point using high-frequency radio signals (typically operating at 2.4 GHz or 5 GHz), removing the requirement for physical cabling.

2. SSID (Service Set Identifier)

The SSID is the broadcast name associated with an 802.11 wireless local area network (WLAN).

Operation: It serves as a unique label that differentiates one wireless network from another in the vicinity. Client devices scan the available SSIDs to locate and join the intended network.

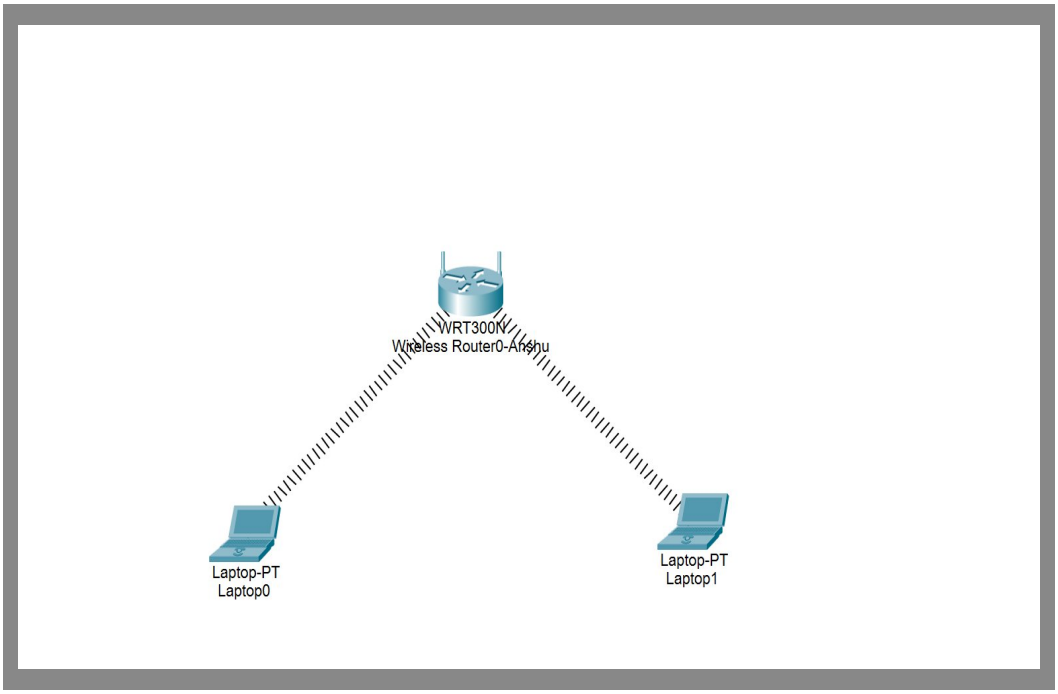
3. WPA2 (Wi-Fi Protected Access 2)

WPA2 is a widely adopted security standard and encryption framework for wireless networks.

Operation: It employs the Advanced Encryption Standard (AES) algorithm to protect transmitted data. The WPA2-PSK (Pre-Shared Key) mode requires a passphrase for authentication, ensuring that only authorised devices can access the network.

Network Topology

A topology was designed using one Wireless Router (WRT300N named "Wireless Router0-Anshu") and two client devices (Laptop-PT named "Laptop0" and "Laptop1"). The Wireless Router serves as the central access point, broadcasting the wireless signal to both laptops.



Configuration

The Wireless Router was set up to function as a DHCP server, automatically distributing IP addresses to all connecting client devices.

IP Configuration Table

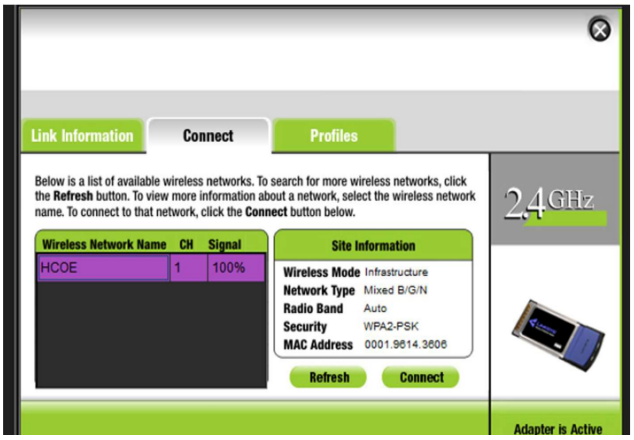
Device	**Interface**	**IP Address**	**Subnet Mask**	**Default Gateway**	**Allocation Method**
Wireless Router0	LAN / Wireless	192.168.0.1	255.255.255.0	-	Static
Laptop0	Wireless0	192.168.0.102	255.255.255.0	192.168.0.1	DHCP
Laptop1	Wireless0	192.168.0.103	255.255.255.0	192.168.0.1	DHCP

Procedure

1. Router Setup: The GUI interface of the Wireless Router was accessed. In "Wireless Settings," the SSID was configured as "HCOE". Under "Wireless Security,"

WPA2-Personal was chosen and the passphrase `hcoe@123` was set.

2. Client Connection: On Laptop0, the "PC Wireless" application was launched and the "Connect" tab was selected. The utility scanned for nearby wireless networks and detected "HCOE" operating on Channel 1 with 100% signal strength.



3. Authentication: The "HCOE" network was selected and the "Connect" button was clicked. A security prompt appeared asking for the Pre-shared Key. The passphrase `hcoe@123` was entered to complete authentication.

4. IP Assignment: After successful authentication, both laptops automatically sent DHCP requests and were assigned IP addresses by the Wireless Router.

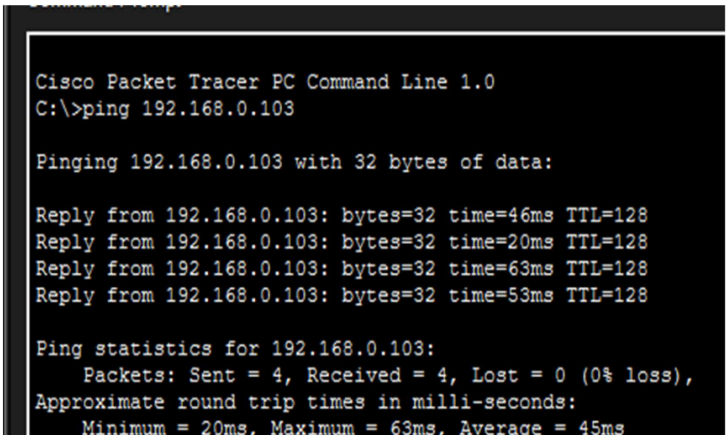
Testing

Connectivity Verification:

To confirm that packets were flowing correctly across the wireless network, a ping test was performed from Laptop0 to Laptop1, which had been assigned the IP address 192.168.0.103.

The Command Prompt on Laptop0 was opened and the command `ping 192.168.0.103` was executed.

We opened the Command Prompt on Laptop0 and typed `ping 192.168.0.103`.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.103

Pinging 192.168.0.103 with 32 bytes of data:

Reply from 192.168.0.103: bytes=32 time=46ms TTL=128
Reply from 192.168.0.103: bytes=32 time=20ms TTL=128
Reply from 192.168.0.103: bytes=32 time=63ms TTL=128
Reply from 192.168.0.103: bytes=32 time=53ms TTL=128

Ping statistics for 192.168.0.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 63ms, Average = 45ms
```

Discussion and Conclusion

In this lab session, a wireless network was successfully configured using a WRT300N router named Wireless Router0-Anshu. The SSID was set to "HCOE" and the network was protected with WPA2-Personal encryption using a pre-shared passphrase. It was observed that both client laptops were able to detect the broadcasted SSID, authenticate using the correct credentials, and obtain dynamic IP addresses via DHCP automatically. The ping test carried out between Laptop0 and Laptop1 returned successful replies with 0% packet loss, confirming full end-to-end wireless connectivity. Thus, the lab was completed with a thorough understanding and practical implementation of Wireless Network Setup and Packet Analysis using Cisco Packet Tracer.